

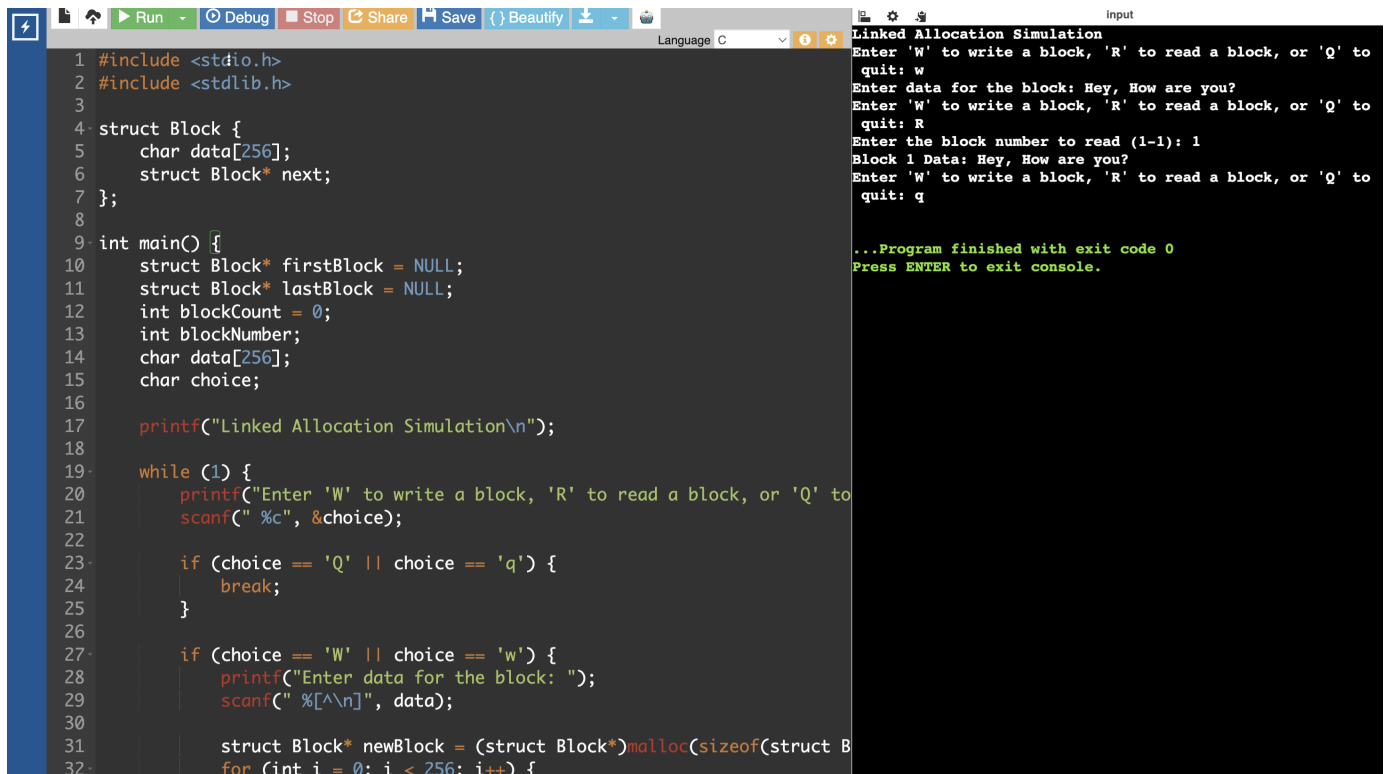
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 36

QUESTION

With linked allocation, each file is a linked list of disk blocks; the disk blocks may be scattered anywhere on the disk. The directory contains a pointer to the first and last blocks of the file. Each block contains a pointer to the next block. Design a C program to simulate the file allocation strategy.

PROGRAM



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Block {
5     char data[256];
6     struct Block* next;
7 };
8
9 int main() {
10     struct Block* firstBlock = NULL;
11     struct Block* lastBlock = NULL;
12     int blockCount = 0;
13     int blockNumber;
14     char data[256];
15     char choice;
16
17     printf("Linked Allocation Simulation\n");
18
19     while (1) {
20         printf("Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: ");
21         scanf("%c", &choice);
22
23         if (choice == 'Q' || choice == 'q') {
24             break;
25         }
26
27         if (choice == 'W' || choice == 'w') {
28             printf("Enter data for the block: ");
29             scanf("%[^\n]", data);
30
31             struct Block* newBlock = (struct Block*)malloc(sizeof(struct Block));
32             for (int i = 0; i < 256; i++) {
```

Linked Allocation Simulation

Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: w

Enter data for the block: Hey, How are you?

Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: R

Enter the block number to read (1-1): 1

Block 1 Data: Hey, How are you?

Enter 'W' to write a block, 'R' to read a block, or 'Q' to quit: q

...Program finished with exit code 0

Press ENTER to exit console.